LAB TASK 11

Implement Matrix Chain Multiplication using Dynamic Programming

CODE:

```
#include <iostream>
#include <vector>
#include <climits>
using namespace std;

int main()
{
   int n;

   cout << "=========================================\n";
   cout << "   MATRIX CHAIN MULTIPLICATION\n";
   cout << "      Dynamic Programming\n";
   cout << "=========================================\n";

   cout << "\nEnter the number of matrices: ";
   cin >> n;

   if (n <= 0)
   {
      cout << "Invalid number of matrices!\n";
      return 0;
   }

   vector<int> p(n + 1);

   cout << "\nEnter the dimensions of matrices:\n";
```

```
for (int i = 0; i <= n; i++)
{
   cout << "Dimension p[" << i << "]: ";
   cin >> p[i];

   if (p[i] <= 0)
   {
      cout << "Dimensions must be positive!\n";
      return 0;
   }
}

cout << "\nMatrix Details:\n";

for (int i = 1; i <= n; i++)
{
   cout << "Matrix A" << i << " : "
        << p[i - 1] << " x " << p[i] << endl;
}

vector<vector<long long>> dp(
   n + 1, vector<long long>(n + 1, 0)
);

for (int len = 2; len <= n; len++)
{
   for (int i = 1; i <= n - len + 1; i++)
   {
      int j = i + len - 1;

      dp[i][j] = LLONG_MAX;

      for (int k = i; k < j; k++)
      {
         long long cost = dp[i][k]
```

```
                        + dp[k + 1][j]
                        + 1LL * p[i - 1] * p[k] * p[j];

                if (cost < dp[i][j])
                {
                    dp[i][j] = cost;
                }
            }
        }
    }

    cout << "\n========================================\n";
    cout << "            FINAL RESULT\n";
    cout << "========================================\n";

    cout << "Total number of matrices: " << n << endl;

    cout << "Minimum number of scalar multiplications: "
        << dp[1][n] << endl;

    cout << "\nThe optimal multiplication cost is "
        << dp[1][n] << ".\n";

    cout << "Dynamic Programming successfully "
        << "minimized the multiplication cost.\n";

    cout << "========================================\n";

    return 0;
}
```

OUTPUT:

```
========================================
   MATRIX CHAIN MULTIPLICATION
      Dynamic Programming
========================================

Enter the number of matrices: 3

Enter the dimensions of matrices:
Dimension p[0]: 10
Dimension p[1]: 30
Dimension p[2]: 5
Dimension p[3]: 60

Matrix Details:
Matrix A1 : 10 x 30
Matrix A2 : 30 x 5
Matrix A3 : 5 x 60

========================================
             FINAL RESULT
========================================
Total number of matrices: 3
Minimum number of scalar multiplications: 4500

The optimal multiplication cost is 4500.
Dynamic Programming successfully minimized the multiplication cost.
========================================
```

Time and Space Complexity

Time Complexity	$O(n^3)$
Space Complexity	$O(n^2)$